Let $f(\underline{x};\theta)$ be the joint pdf of $X = (X_1, \ldots, X_n)$ so the log likelihood is

$$\ell(\theta;\underline{x}) = \log f(\underline{x};\theta) \qquad \theta \in \Theta$$

(Fisher) Score function is

$$U(\theta) = U(\theta;\underline{x}) = \frac{\partial}{\partial\theta}\ell(\theta;\underline{x}) \qquad \theta \in \Theta$$

Assuming we can interchange differentiation wrt θ and integration wrt x twice, so

$$\int \frac{\partial}{\partial\theta} f(\underline{x};\theta)\,d\underline{x} = \frac{\partial}{\partial\theta}\int f(\underline{x};\theta)\,d\underline{x}$$

Then $\mathbb{E}_\theta[U(\theta;X)] = \mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(\underline{X};\theta)\right]$

CB eq 7.3.7-8 $= \int \frac{1}{f(\underline{x};\theta)}\left\{\frac{\partial}{\partial\theta} f(\underline{x};\theta)\right\} f(\underline{x};\theta)\,d\underline{x}$

$$= \int \frac{\partial}{\partial\theta} f(\underline{x};\theta)\,d\underline{x} = \frac{\partial}{\partial\theta}\int f(\underline{x};\theta)\,d\underline{x} = \frac{\partial}{\partial\theta}[1] = 0$$

The quantity $\mathrm{var}_\theta[U(\theta;X)] = \mathbb{E}_\theta[U^2(\theta;X)] =: I(\theta)$ is called the expected (Fisher) information.

Thursday 13 August (1)

Assuming we can interchange differentiation wrt θ and integration wrt $\underline{x}$ twice, that is ($d\underline{x} = dx_1, \ldots dx_n$)

$$\int \frac{\partial^2}{\partial\theta^2} f(\underline{x};\theta)\,d\underline{x} = \frac{\partial^2}{\partial\theta^2}\int f(\underline{x};\theta)\,d\underline{x}$$

Note that

$$\frac{\partial^2}{\partial\theta^2}\log f(\underline{x};\theta) = \frac{\partial}{\partial\theta}\left\{\frac{1}{f(\underline{x};\theta)}\frac{\partial}{\partial\theta} f(\underline{x};\theta)\right\}$$

$$= \frac{-1}{f(\underline{x};\theta)^2}\left\{\frac{\partial}{\partial\theta} f(\underline{x};\theta)\right\}^2 + \frac{1}{f(\underline{x};\theta)}\frac{\partial^2}{\partial\theta^2} f(\underline{x};\theta)$$

$$I(\theta) = \mathbb{E}_\theta\left[U(\theta;\underline{X})^2\right] =$$

$$= \mathbb{E}_\theta\left[\frac{1}{f(\underline{x};\theta)}\left\{\frac{\partial^2}{\partial\theta^2} f(\underline{x};\theta)\right\} - \frac{\partial^2}{\partial\theta^2}\log f(\underline{x};\theta)\right]$$

$$= \int\left\{\frac{1}{f(\underline{x};\theta)}\left\{\frac{\partial^2}{\partial\theta^2} f(\underline{x};\theta)\right\} - \frac{\partial^2}{\partial\theta^2}\log f(\underline{x};\theta)\right\} f(\underline{x};\theta)\, d\underline{x}$$

$$= \int \frac{\partial^2}{\partial\theta^2} f(\underline{x};\theta)\, d\underline{x} - \int\left\{\frac{\partial^2}{\partial\theta^2}\log f(\underline{x};\theta)\right\} f(\underline{x};\theta)\, d\underline{x}$$

$$= \frac{\partial^2}{\partial\theta^2}\int f(\underline{x};\theta)\, d\underline{x} - \int\left\{\frac{\partial^2}{\partial\theta^2}\log f(\underline{x};\theta)\right\} f(\underline{x};\theta)\, d\underline{x}$$

$$= -\int\left\{\frac{\partial^2}{\partial\theta^2}\log f(\underline{x};\theta)\right\} f(\underline{x};\theta)\, d\underline{x}$$

$$= -\mathbb{E}_\theta\left(\frac{\partial^2}{\partial\theta^2}\log f(\underline{x};\theta)\right) \quad \text{(CB Lemma 7-3-11)}$$

The negative of the second derivative of the log likelihood is the observed information

$$J(\theta) = -\frac{\partial^2}{\partial\theta^2}\ell(\theta) \qquad \text{note } I(\theta) = \mathbb{E}_\theta J(\theta)$$

When $X_1, \ldots, X_n$ is a random sample of size n (that is independent and identically distributed), the likelihood is

$$L(\theta;\underline{x}) = \prod_{j=1}^{n} f(x_j;\theta)$$

← marginal pdf of a single observation.

and the log likelihood is $\ell(\theta) = \sum_{j=1}^{n}\log f(x_j;\theta)$

So the score function is $U(\theta) = \sum_{j=1}^{n}\frac{\partial}{\partial\theta}\log f(x_j;\theta)$

and the expected information

$$I(\theta) = \mathbb{E}\,\mathrm{Var}_\theta\left[U(\theta;\underline{X})\right] = n\,\mathrm{Var}_\theta\left(\underbrace{\frac{\partial}{\partial\theta}\log f(X;\theta)}_{\text{score of single obs.}}\right) = n\,i(\theta)$$

Example (Bernoulli) Suppose $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$. Then the log likelihood is

$$\ell(p) = s \log(p) + (n-s)\log(1-p) \quad \text{where } s = \sum_{j=1}^{n} x_j.$$

The score function is

$$U(p; \underline{x}) = \frac{\partial}{\partial p}\ell(p) = \frac{s}{p} - \frac{(n-s)}{1-p}$$

$$= \frac{s - np}{p(1-p)}$$

The expected information is

$$S = \sum_{j=1}^{n} X_j$$

$$I(p) = \text{Var}_p\left(U(p; \underline{X})\right) = \text{Var}_p\left(\frac{S - np}{p(1-p)}\right)$$

$$= \frac{\text{Var}_p(S)}{p^2(1-p)^2} = \frac{np(1-p)}{p^2(1-p)^2} = \frac{n}{p(1-p)}$$

note $i(\theta) = \frac{1}{p(1-p)}$.

The observed information is

$$J(p) = -\frac{\partial^2}{\partial p^2}\ell(p) = \frac{-\partial}{\partial p}\left\{\frac{s - np}{p(1-p)}\right\} = -\frac{(-np(1-p) - (1-2p)(s-np))}{p^2(1-p)^2}$$

Cramer - Rao lower bound (CB§ Thm 7.3.9, cor 7.3.10)

Let $T(\underline{X})$ be an unbiased estimator of $g(\theta)$, that is

$$\int T(\underline{x}) f(\underline{x};\theta)\, d\underline{x} = \mathbb{E}_\theta\, T(\underline{X}) = g(\theta).$$

Differentiate both sides (above) wrt θ, and assume we can interchange differentiating wrt θ and integration wrt $\underline{x}$

$$g'(\theta) = \frac{\partial}{\partial\theta} \int T(\underline{x})\, f(\underline{x};\theta)\, d\underline{x}$$

$$= \int T(\underline{x}) \left\{ \frac{\partial}{\partial\theta} f(\underline{x};\theta) \right\} d\underline{x}$$

$$= \int T(\underline{x}) \frac{1}{f(\underline{x};\theta)} \left\{ \frac{\partial}{\partial\theta} f(\underline{x};\theta) \right\} f(\underline{x};\theta)\, d\underline{x}$$

$$= \mathbb{E}_\theta \left[T(\underline{X}) \cdot U(\theta;\underline{X}) \right]$$

Since $\mathbb{E}_\theta[U(\theta;\underline{X})] = 0$ $\qquad \operatorname{cov}(V,W) = \mathbb{E}(VW) - \mathbb{E}(V)\mathbb{E}(W)$

$$\mathbb{E}_\theta \left[T(\underline{X}) \cdot U(\theta;\underline{X}) \right] = \operatorname{cov}_\theta \left(T(\underline{X}), U(\theta;\underline{X}) \right)$$

Recall $\operatorname{cov}(V,W)^2 \leq \operatorname{Var}(V) \operatorname{Var}(W)$ STAT2003 claim 38 cf cauchy-schwarz inequality

So $$\left(g'(\theta)\right)^2 \leq \operatorname{Var}_\theta\left(T(\underline{X})\right) \cdot \underbrace{\operatorname{Var}_\theta\left(U(\theta;\underline{X})\right)}_{I(\theta)}$$

Rearranging we get

$$\operatorname{Var}_\theta\left(T(\underline{X})\right) \geq \frac{\left(g'(\theta)\right)^2}{I(\theta)}$$

In the case of IID observations $\operatorname{Var}_\theta(T(\underline{X})) \geq \dfrac{\left(g'(\theta)\right)^2}{n\, i(\theta)}$

Example (Bernoulli) $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$. Let $T(X_1, \ldots, X_n)$ be any unbiased estimator of $g(p) = p$. Then $g'(p) = 1$ and the Cramer-Rao lower bound becomes

$$\text{Var}_p\left(T(X_1, \ldots, X_n)\right) \geq \frac{1^2}{n/p(1-p)} = \frac{p(1-p)}{n}.$$

The unbiased estimator $\bar{X} = \frac{1}{n}\sum_{j=1}^{n} X_j$ achieves this lower bound.

(When does an estimator achieving the lower bound exist? CB cor 7.3.15)

Regularity Assumptions (CB 10.6.2)

- $X_1, \ldots, X_n$ is a random sample from $f(x; \theta_0)$ (i.e. the $X_1, \ldots, X_n$ are IID).
- The model is identifiable — distinct values of θ correspond to distinct distributions.
- The pdfs/pmfs $f(x;\theta)$ have the same support for all $\theta \in \Theta$.
- The parameter space Θ is an open set.

Further assumptions concerning the interchange of derivatives and integrals, and "smoothness" of $f(\cdot;\theta)$.

Consistency of MLEs (CB Thm 10.1.6)

Recall a sequence of estimators $T_n(X_1, \ldots, X_n)$ is consistent estimator for $\psi(\theta)$ if $T_n(X_1, \ldots, X_n) \xrightarrow{P} \psi(\theta)$, i.e.

$$\forall \varepsilon > 0 \quad \lim_{n\to\infty} \mathbb{P}\left(|T_n(X_1, \ldots, X_n) - \psi(\theta)| > \varepsilon\right) = 0.$$

Let $X_1, \ldots, X_n \sim f(x; \theta_0)$ (true value) and $\hat{\theta}$ be the MLE of θ. Let $g: \Theta \to \mathbb{R}$ be a continuous function. Then $g(\hat{\theta}) \xrightarrow{P} g(\theta_0)$.

Sketching ideas of proof: Note that if $\hat{\theta} \xrightarrow{p} \theta_0$, then $g(\hat{\theta}) \xrightarrow{p} g(\theta_0)$ by continuous mapping theorem (Wk 1, lec 2)
By the law of large numbers

$$n^{-1} \sum_{j=1}^{n} \log f(x_j;\theta) \xrightarrow{P} \mathbb{E}_{\theta_0}\left[\log f(x;\theta)\right]$$

It can be shown that

$$\mathbb{E}_{\theta_0}\left[\log f(x;\theta)\right] - \mathbb{E}_{\theta_0}\left[\log f(x;\theta_0)\right] = \mathbb{E}_{\theta_0}\left[\log\left(\frac{f(x;\theta)}{f(x;\theta_0)}\right)\right] < 0$$

for all $\theta \neq \theta_0$ by Jensen's inequality (which says $\mathbb{E}\, g(Y) > g(\mathbb{E}\, Y)$ for g convex)

Efficiency of MLEs

Def: For an estimator $T_n(X_1, \ldots, X_n)$ of ψ. Suppose $k_n \uparrow \infty$ such that $k_n(T_n - \psi) \xrightarrow{d} N(0, \sigma^2)$. Then σ^2 is called the asymptotic variance of T_n (RB Def 10.1.9)

This is not necessarily the same as $\lim_{n\to\infty} k_n^2 \operatorname{Var}(T_n)$.
Eg. suppose $Z \sim N(0,1)$ and $Y \sim t_2$, and let $T_n = Z + \frac{1}{n} Y$.
Then $\operatorname{Var}(T_n) = \infty$ but $T_n \xrightarrow{d} N(0,1)$ so the asymptotic variance of T_n is 1.

Def: A sequence of estimators W_n is asymptotically efficient for $g(\theta)$ if

$$\sqrt{n}\,(W_n - g(\theta)) \to N(0, v(\theta))$$

where $$v(\theta) = \frac{(g'(\theta))^2}{i(\theta)}$$

$i(\theta)$: information of single observation.

Theorem: Let $X_1, \ldots, X_n \sim f(\cdot;\theta_0)$ and g a continuous function on Θ. Under regularity assumptions,

$$\sqrt{n}\,(g(\hat{\theta}) - g(\theta_0)) \xrightarrow{d} N(0, V(\theta))$$

where

$$V(\theta) = \frac{(g'(\theta))^2}{i(\theta)}.$$

In practice, for "sufficiently large n" the distribution of $\hat{\theta}$ is approximately

$$\hat{\theta} \sim N\left(\theta_0, \frac{1}{n i(\theta_0)}\right)$$

replace θ_0 by $\hat{\theta}$.

and

$$g(\hat{\theta}) \sim N\left(g(\theta_0), \frac{(g'(\theta_0))^2}{n i(\theta_0)}\right)$$

Sketch ideas of proof: Expanding the derivative of the log likelihood as a Taylor series about θ_0.

$$l'(\theta; \underline{x}) = l'(\theta_0; \underline{x}) + (\theta - \theta_0)\, l''(\theta_0; \underline{x}) + \ldots$$

ignore higher order terms

At the MLE $\hat{\theta}$, $l'(\hat{\theta}; \underline{x}) = 0$ so

$$0 = l'(\theta_0; \underline{x}) + (\hat{\theta} - \theta_0)\, l''(\theta_0; \underline{x})$$

Rearranging

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{\sqrt{n}\, l'(\theta_0; \underline{x})}{-l''(\theta_0; \underline{x})} = \frac{n^{-1/2}\, l'(\theta_0; \underline{x})}{-\frac{1}{n} l''(\theta_0; \underline{x})}$$

By CLT $\quad n^{-1/2}\, l'(\theta_0; \underline{X}) \xrightarrow{d} N(0, i(\theta_0))$.

By LLN $\quad -\frac{1}{n} l''(\theta_0; \underline{X}) \xrightarrow{P} i(\theta_0)$.

$$f(x;\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0,1,2,\ldots \quad \lambda \geq 0$$

Example (Poisson) $X_1,\ldots,X_n \sim \text{Poisson}(\lambda)$. We want to estimate $\mathbb{P}(X=0) = e^{-\lambda}$.

Likelihood function $\mathcal{L}(\lambda;\underline{x}) = \prod_{j=1}^{n}\left(\frac{e^{-\lambda}\lambda^{x_j}}{x_j!}\right) = \frac{e^{-n\lambda}\lambda^{s}}{\prod_{j=1}^{n} x_j!}$

where $s = \sum_{j=1}^{n} x_j$

log likelihood function $l(\lambda;\underline{x}) = -n\lambda + s\log\lambda - \sum_{j=1}^{n}\log(x_j!)$

The score function $U(\lambda;\underline{x}) = l'(\lambda;\underline{x}) = -n + \frac{s}{\lambda}$

So the MLE is $\hat{\lambda} = s/n$ (ie $\bar{x}$) should still check we have maximum.

Expected information $I(\lambda) = \text{Var}_\lambda\left(U(\lambda;\underline{X})\right) = \text{Var}_\lambda\left(-n + \frac{S}{\lambda}\right)$

where $S = \sum_{j=1}^{n} X_j$. $= \frac{\text{Var}_\lambda(S)}{\lambda^2} = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda}$.

(The observed information $J(\lambda) = -\frac{\partial^2 l}{\partial\lambda^2}(\lambda) = \frac{+S}{\lambda^2}$

We want to estimate $e^{-\lambda}$. The MLE of $e^{-\lambda}$ is $e^{-\bar{X}}$.

The asymptotic variance is

$$\frac{(g'(\lambda))^2}{I(\lambda)} = \frac{(-e^{-\lambda})^2}{1/\lambda} = \lambda e^{-2\lambda}$$

So $\text{Var}(e^{-\bar{X}}) \simeq \frac{\lambda e^{-2\lambda}}{n}$.

Compare to the estimator $\bar{Y} = \frac{1}{n}\sum_{j=1}^{n} Y_j$ where $Y_j = 1$ if $X_j = 0$ and $Y_j = 0$ if $X_j > 0$. So $Y_j \sim \text{Bernoulli}(e^{-\lambda})$.

So $\bar{Y}$ is unbiased estimator of $e^{-\lambda}$, and

$$\text{Var}_\lambda(\bar{Y}) = \frac{e^{-\lambda}(1-e^{-\lambda})}{n}$$

Comparing with CR lower bound $\frac{e^{-\lambda}(1-e^{-\lambda})}{n} \geq \frac{\lambda e^{-2\lambda}}{n}$

$\Rightarrow e^{\lambda} - 1 \geq \lambda$?